

Co-factors and defining determinants

Definition

- 1 Suppose $A = (a_{ij})$ is an $n \times n$ matrix. Then the submatrix A_{ij} is obtained by removing the i^{th} row and j^{th} column from A . The (i, j) -cofactor is

$$C_{ij} := (-1)^{i+j} \det(A_{ij}).$$

- 2 We define the determinant inductively: if A is an $n \times n$ matrix and $1 \leq i \leq n$ then

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}.$$

expansion
across
a row

Equivalently, for any $1 \leq j \leq n$,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}.$$

exp. down
a column.

Ex: $\det \begin{pmatrix} 2 & 1 & 4 \\ 1 & 0 & -1 \\ 3 & 2 & 0 \end{pmatrix}$

Compute the determinant.

$$= - \det \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 6 \\ 0 & 2 & 3 \end{pmatrix} \begin{matrix} \leftarrow -2 \times R_1 + R_2 \\ \leftarrow -3 \times R_1 + R_3 \end{matrix}$$

$$= - \det \begin{pmatrix} 1 & 6 \\ 2 & 3 \end{pmatrix} \\ = - (3 - 12) \\ = 9.$$

$$= - \det \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 6 \\ 0 & 0 & -9 \end{pmatrix} \leftarrow -2R_2 + R_3$$

$$= - (1)(1)(-9) = 9$$

Properties of the determinant

Theorem

Suppose that B is obtained from A by a single row operation.

- 1 If the row operation is interchanging two rows the $\det(B) = -\det(A)$.*
- 2 If the row operation is multiplying a row by k then $\det(B) = k \det(A)$.*
- 3 If the row operation is adding a multiple of one row to another then $\det(B) = \det(A)$.*

Theorem

For any A , $\det(A) = \det(A^T)$.



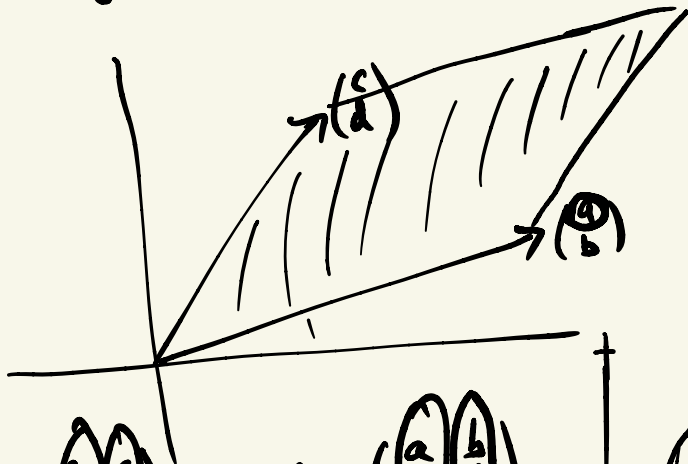
Ex: $\text{Det} \begin{pmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$ ← matrix from Monday

$$= \text{Det} \begin{pmatrix} 1 & 0 & 0 \\ 3 & 4 & -2 \\ 2 & 2 & -3 \end{pmatrix}$$

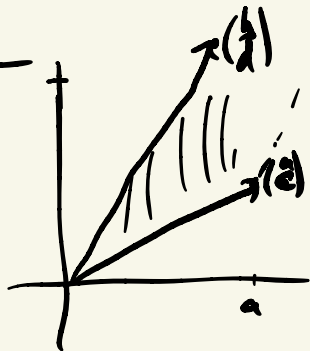
$\uparrow$ $\uparrow$
 $C1 + C2$ $-2C1 + C3$

$$= 1 \cdot \det \begin{pmatrix} 4 & -2 \\ 2 & -3 \end{pmatrix} = -12 - (-4) = -8$$

Ex of geometric meaning of $\det(A) = \det(A^T)$



$$\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$



Properties of the determinant

Theorem

$$\det(AB) = \det(A) \det(B).$$

Theorem

$\det(A) \neq 0$ iff A is invertible.

If A is invertible then $\text{ref}(A)$ is I_n .

$\det(I_n) = 1$. Since all elementary ops change the deter. by a non-zero amount
 $\det(A) \neq 0$

Properties of the determinant

Theorem

If $v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n$ are $(n-1)$ vectors in $\mathbb{R}^n$ then the map $T : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$T(x) = \det(v_1, \dots, v_{j-1}, x, v_{j+1}, \dots, v_n)$$

is a linear transformation.

Handwritten diagram illustrating the map T . The determinant is written as $\det \begin{pmatrix} | & | & \dots & | & | & \dots & | \\ v_1 & v_2 & \dots & v_{j-1} & x & v_{j+1} & \dots & v_n \\ | & | & \dots & | & | & \dots & | \end{pmatrix}$. A circle is drawn around the column x . An arrow points from this circle to the expression $x \mapsto \det(\)$.

$$\begin{aligned}
 T(u+v) &= \det(v_1, v_2, \dots, \underbrace{u+v}_{j \text{ col.}}, \dots, v_n) \\
 &= \det(v_1, \dots, \overset{\uparrow}{u} \dots v_n) + \det(v_1, \dots, \underset{\downarrow}{v} \dots v_n) = T(u) + T(v)
 \end{aligned}$$

$$\begin{aligned}
 T(ku) &= \det(v_1, v_2, \dots, \overset{\uparrow}{ku} \dots v_n) \\
 &= k \det(v_1, \dots, u \dots v_n) \\
 &= k T(u)
 \end{aligned}$$

Ex:

$$\det \begin{pmatrix} 1 & 0 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{pmatrix} = \det \begin{pmatrix} \cancel{1} & 0 & -1 \\ \cancel{0} & 3 & 2 \\ \cancel{0} & 0 & 4 \end{pmatrix}$$

$$\begin{aligned} &= \\ &12 \end{aligned}$$

$$= \det \begin{pmatrix} \cancel{1} & 0 & -1 \\ \cancel{0} & 3 & 2 \\ \cancel{0} & 0 & 4 \end{pmatrix} + \det \begin{pmatrix} \cancel{1} & 0 & -1 \\ \cancel{0} & 3 & 2 \\ \cancel{0} & 0 & 4 \end{pmatrix}$$

$\begin{matrix} C_1 & & \\ \downarrow & & \end{matrix}$

$\begin{matrix} = \\ 12 \end{matrix}$ $\begin{matrix} = \\ 0 \end{matrix}$

$$\det \begin{pmatrix} -1 & 0 & -1 \\ 2 & 3 & 2 \\ 4 & 0 & 4 \end{pmatrix} = 3 \cdot \det \begin{pmatrix} -1 & -1 \\ 4 & 4 \end{pmatrix}$$
$$= 3(-4 - (-4)) = 0$$

Cramer's rule

Definition

For an $n \times n$ matrix A and any vector $b \in \mathbb{R}^n$, we define $\det(A)_i(b)$ as the determinant of the matrix A with the i^{th} column replaced by b .

Theorem (Cramer's rule)

If A is invertible then the solution of $Ax = b$ is given by

$$x_i = \frac{\det(A)_i(b)}{\det(A)} \text{ for } i \leq n.$$